Schema chapter 3

The following tables form part of a database held in a relational DBMS:

Hotel	(hotelNo, hotelName, city)
Room	(roomNo, hotelNo, type, price)
Booking	(hotelNo, guestNo, dateFrom, dateTo, roomNo)
Guest	(guestNo, guestName, guestAddress)

where Hotel contains hotel details and hotelNo is the primary key;
Room contains room details for each hotel and (roomNo, hotelNo) forms the primary key;
Booking contains details of bookings and (hotelNo, guestNo, dateFrom) forms the primary key;
Guest contains guest details and guestNo is the primary key.

soal

Answer the following questions using the relational schema from the Exercises at the end of Chapter 3:

6.10 Create the Hotel table using the integrity enhancement features of SQL.

6.11 Now create the Room, Booking, and Guest tables using the integrity enhancement features of SQL with the following constraints:

(a) type must be one of Single, Double, or Family.
(b) price must be between £10 and £100.
(c) roomNo must be between 1 and 100.
(d) dateFrom and dateTo must be greater than today's date.

(e) The same room cannot be double-booked.
(f) The same guest cannot have overlapping bookings.

Table hotel

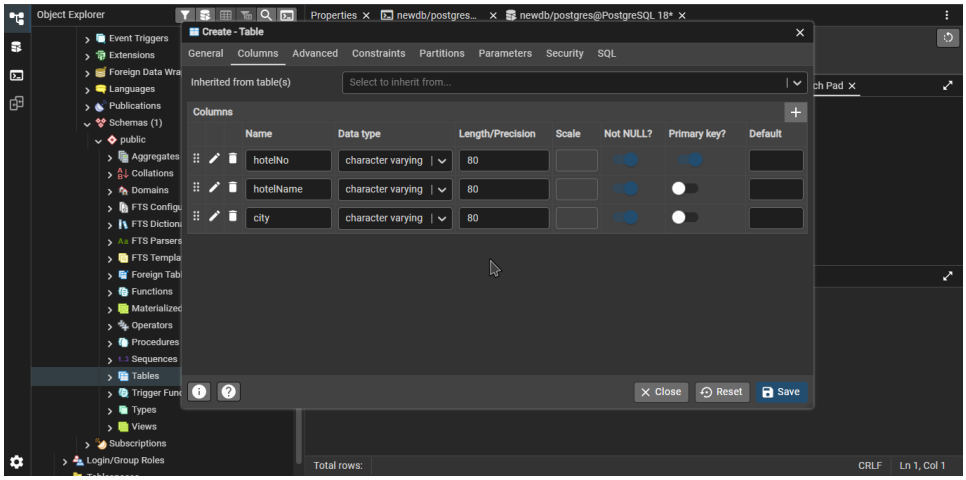
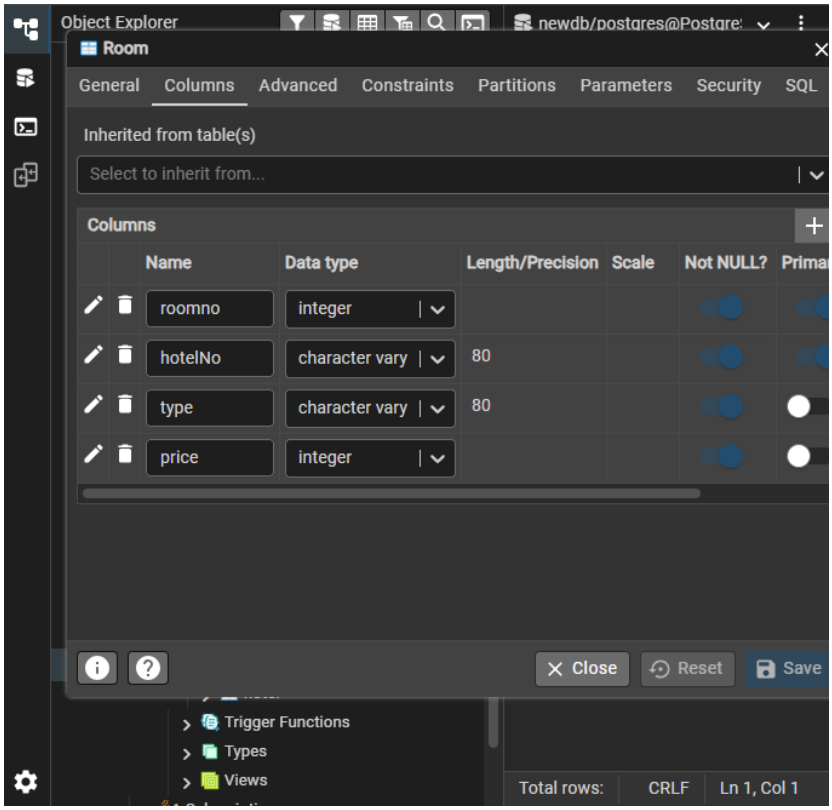


Table room



Check constraint

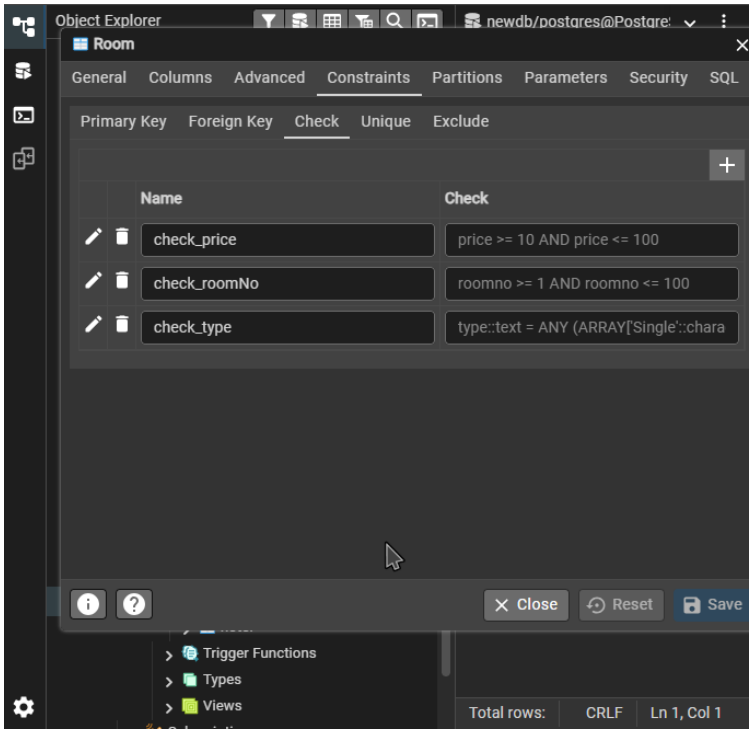
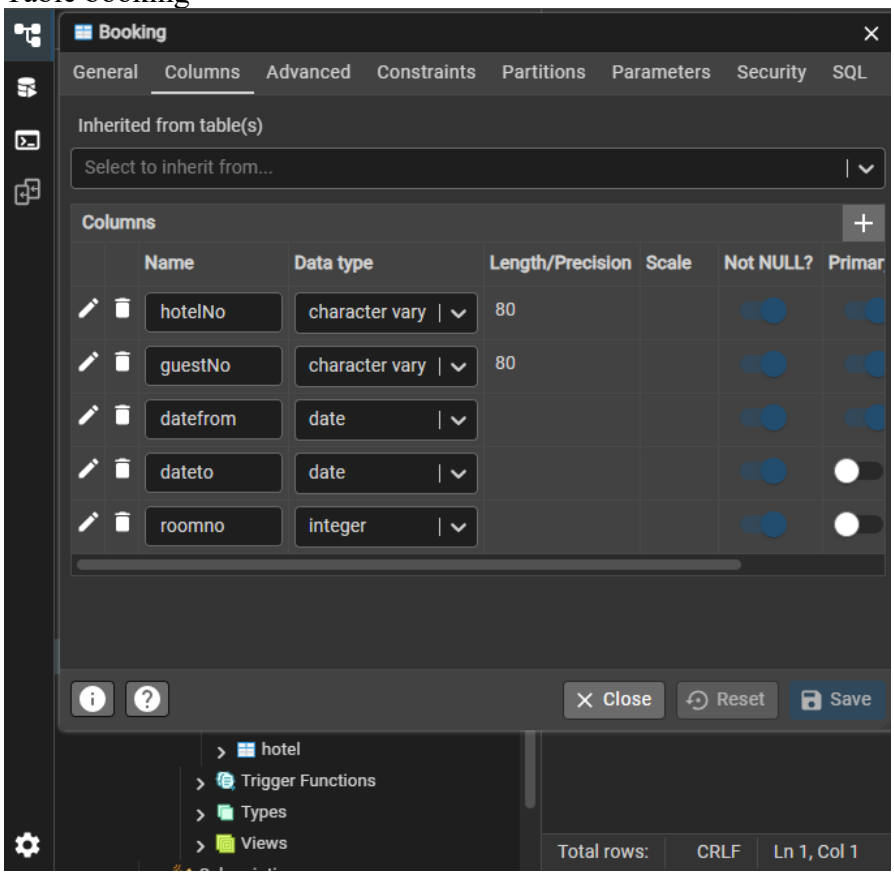


Table booking



Check constraint

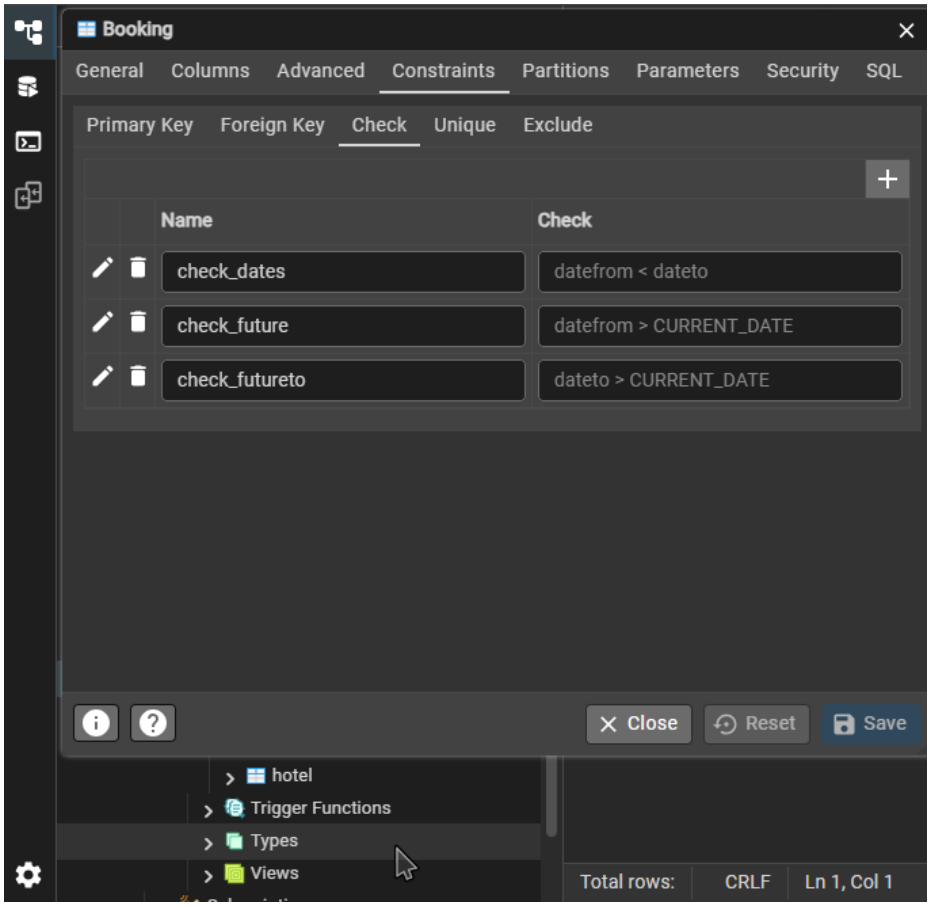
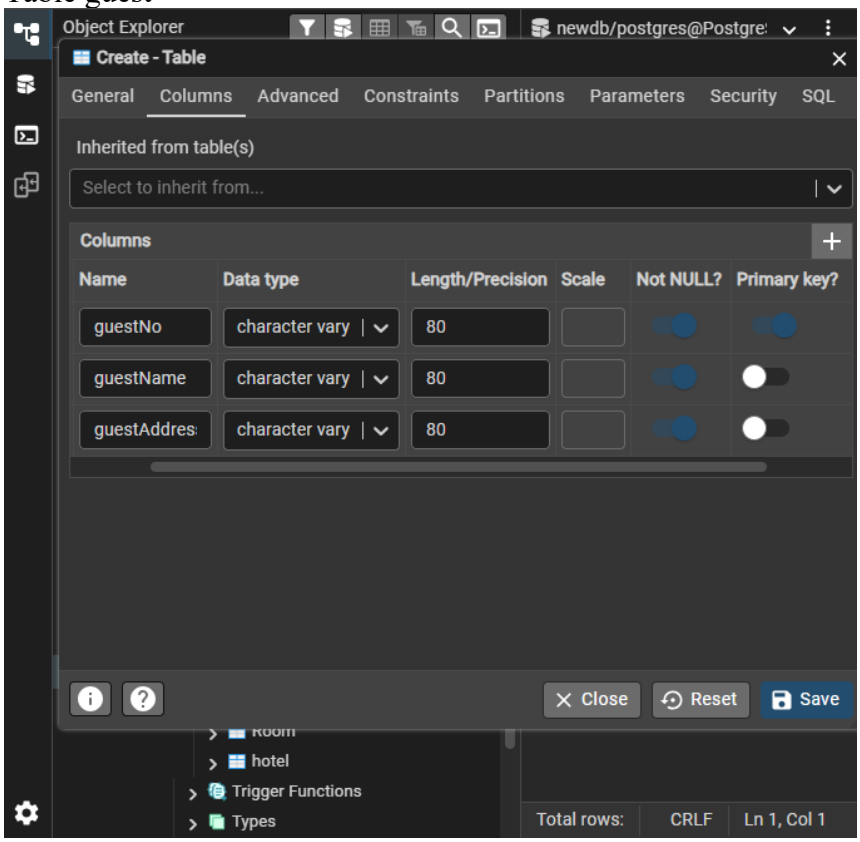
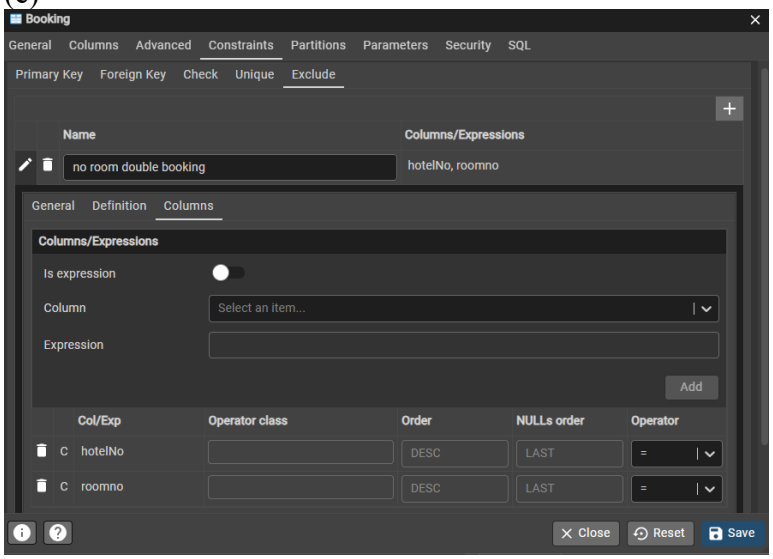
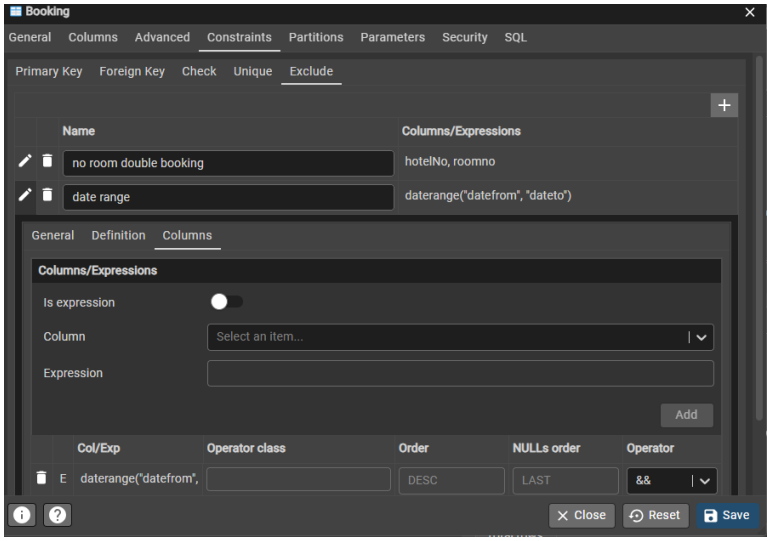


Table guest

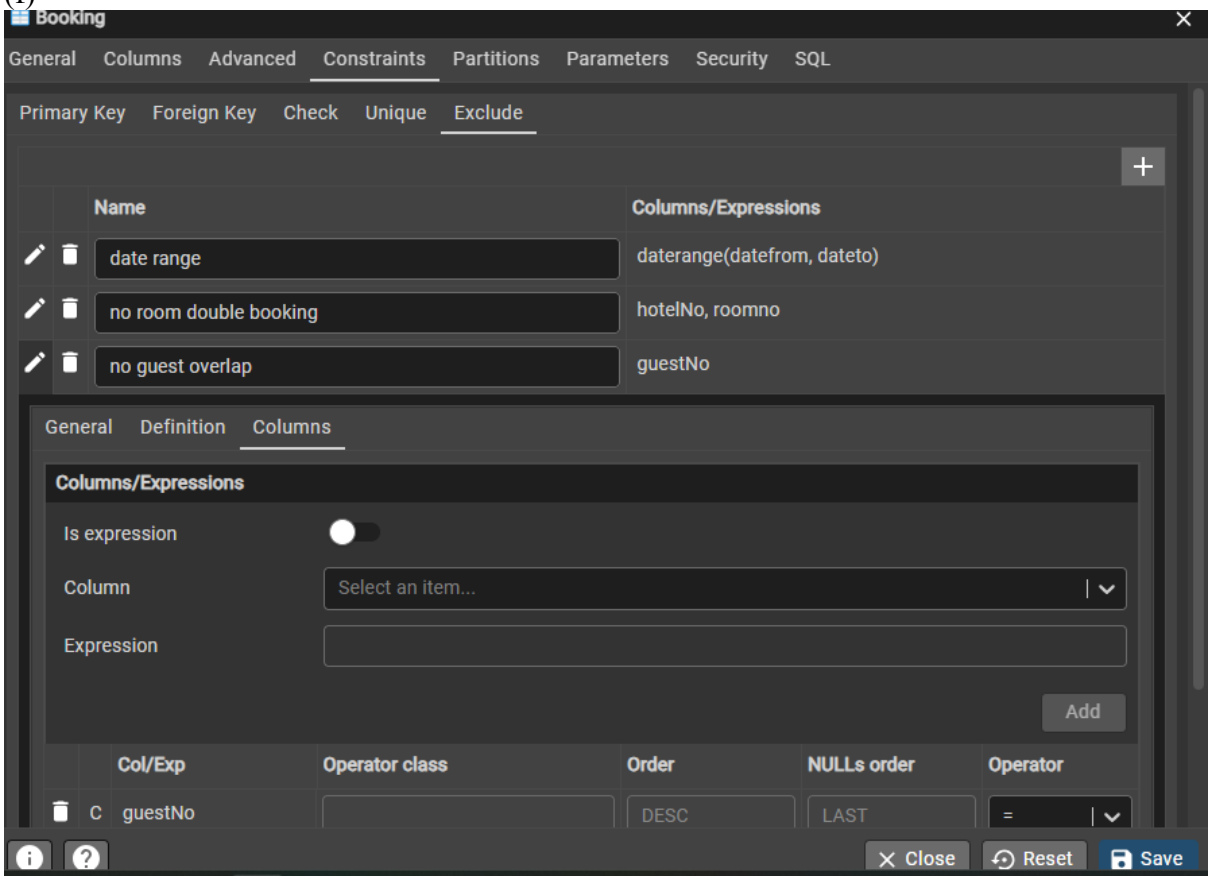


(e)




Booking
General Columns Advanced Constraints Partitions Parameters Security SQL
Primary Key Foreign Key Check Unique Exclude
Name
Columns/Expressions
no room double booking
hotelNo, roomno
date range
daterange("datefrom", "dateto")
General Definition Columns
Columns/Expressions
Is expression
Column
Select an item...
Expression
Add
Col/Exp
Operator class
Order
NULLs order
Operator
E
daterange("datefrom",
DESC
LAST
&&
Close
Reset
Save

(f)


Booking
General Columns Advanced Constraints Partitions Parameters Security SQL
Primary Key Foreign Key Check Unique Exclude
Name
Columns/Expressions
date range
daterange(datefrom, dateto)
no room double booking
hotelNo, roomno
no guest overlap
guestNo
General Definition Columns
Columns/Expressions
Is expression
Column
Select an item...
Expression
Add
Col/Exp
Operator class
Order
NULLs order
Operator
C
guestNo
DESC
LAST
=
Close
Reset
Save